

SYNOPSIS



At 1320, on 19 June 2005, a 13 year old boy suffered serious head injuries when the powerboat in which he was co-driver was struck by another race boat during a K-200 class Junior Offshore National Championship race at Portland Harbour.

The collision occurred as a fleet of 9 boats, each with a maximum speed of about 31mph, was rounding the first 90° left turn marker buoy of an "M" shaped course. The boats were still bunched as the first mark was only 600 metres from the start line. The injured boy's boat, which was being driven by a 10 year old boy who was in his first season of powerboat racing, "hooked" and came to a sudden and unexpected stop directly in the path of a following boat. The other boat, which was being driven by an 11 year old who was also in his first season of racing, had no time in which to take avoiding action. Its bow struck the port side of the stopped boat at or near right angles and, due to its momentum, over rode its side, striking the co-driver who was in the left-hand seat. It then continued to rise vertically into the air, ejecting its two crew into the sea, before coming down stern-first into the water.

The race was stopped very quickly, and first-aid was administered to the injured boy by a paramedic who was a member of the race supervision team. Due to the seriousness of the boy's injuries, he was airlifted by a coastguard helicopter to a local hospital, where he was found to be in a coma. After the collision, the injured boy's safety helmet was retrieved from the sea a short distance away.

K-200 racing involved head-to-head racing by children aged 8 to 16 years of age, divided into two age groups, driving Sorcerer 373 or 375R boats equipped with a 25hp outboard engine. K-class, or junior offshore powerboat racing, had been started by Wanda Offshore Racing Klubb (WORK) 3 years before the accident. WORK is affiliated to the Royal Yachting Association (RYA), which is the governing body for all offshore powerboat racing in the UK, including K-class. The RYA produces an annual racing handbook detailing the rules, regulations and racing calendar for all classes of offshore powerboat racing. The RYA also approves the race officials, requires the individual clubs to send them accident reports, and generally oversees the governance of the sport. This is largely achieved by way of a hierarchical committee/working group system comprising volunteer members who are usually drawn from the powerboating community.

WORK did not carry out detailed risk assessments for the new class, and the interaction of factors like the course design, type of boat to be used, race rules and training requirements were not fully considered. The RYA was not pro-active in monitoring WORK's safety management and development of the class. K-class racing was innovative in that organised junior offshore powerboat racing did not exist previously anywhere in the world.

There were no minimum training requirements to qualify children to participate in K-class racing, although many people believed that 40 hours tuition had been stated and agreed. The children were trained by members of WORK mostly during the winter months and, once judged competent by the instructors and passed fit to race, they were issued with a personal racing licence by the RYA.

The amount of training given to beginners had apparently reduced during the 3 years the class had been in existence. The injured co-driver was an experienced racer. However, the two drivers and the other co-driver, who were directly involved in the accident, were all new entrants for the 2005 season, and had all been signed off as 'ready to race' after receiving fewer than 10 hours tuition. It is believed this inexperience and lack of training had a significant bearing on the circumstances of the accident.

The 3.6 metre Sorcerer boat type was originally designed in the 1960s as a ski boat, with turtle decks and low freeboard to enable skiers to be easily pulled from the water. The boat was widely raced with various sizes of outboard engine and with a great deal of success during the 1970s and 1980s. Production of the boat ceased in 1985. The original Sorcerer 373 and the new 375R were chosen by WORK because they were a good size, provided a stable platform for children to race and there were plenty of cheap secondhand boats available. During the development of the racing class, little attention was paid to the fact that a child would be left unprotected in the event of two boats colliding and one overriding the other.

The value of accident and incident reports was not fully recognised by either WORK or the RYA, and minimal attention was paid to them by both organisations. Some reports, which were required by the RYA for K-class racing, were found to be poorly completed and, in some cases, were received weeks after the RYA's 7 day deadline. The fact that no reports were received from the 2004 racing season went unnoticed at the RYA. The RYA also paid little attention to a letter and enclosed safety assessment which were sent to the organisation in 2004 by the boat's original designer, in which he warned of the dangers associated with the sort of accident that occurred on 19 June 2005.

Better attention to good safety management practices by WORK and the RYA, both during the setting up and development of the class, might well have prevented the accident.

During examination by the MAIB, the dislodged helmet worn by the co-driver was found to have no apparent impact damage. The chin strap was fastened, and the inflatable bladder, which ensured a comfortable fit, was partially inflated, however a sleeve on the chin strap was found to be missing. The MK10 Gecko Marine Safety Helmet was worn by the majority of K-class competitors, and some adults in other powerboat racing classes. It had been designed and manufactured by Gecko Head Gear Ltd, initially for the Royal National Lifeboat Institution (RNLI) to protect its boat crews. It had not been the subject of a detailed assessment to determine its suitability in powerboat racing.

It was also found that, despite a widespread belief that the helmet was “one size fits all”, and therefore appropriate for use by children, the helmet had only been tested in line with the British Standards Institute specification for an adult male head size. The protection afforded by the helmet to persons with head sizes outside this range had therefore not been established.

During the investigation, MAIB issued Safety Bulletin 3/2005, which strongly advised all users of marine safety helmets to ensure helmets are of the correct fit and properly worn at all times. It also recommended that the RYA should conduct a risk assessment to identify a safe helmet to be used by all powerboat racers.

Separate recommendations have now also been made to WORK and the RYA regarding the need for the adoption of good safety management and risk assessment practices in order to help ensure the future safety of the children competing in K-class racing.

Figure 1



Photograph of powerboat K11

SECTION 1 - FACTUAL INFORMATION

1.1 PARTICULARS OF POWERBOATS AND ACCIDENT (Figure 1)

Vessel details

Boat type	:	Sorcerer 373
Length overall	:	3.65 metres/12 feet
Engine power and/or type	:	Mercury 25hp outboard engine
Maximum speed	:	31mph/27knots
Weight of boat and engine	:	Minimum 235kg, without fuel

Accident details

Description	:	Collision between two powerboats during a junior racing event
Time and date	:	1320 on 19 June 2005
Location of incident	:	Portland Harbour
Persons on board	:	2
Injuries/fatalities	:	A 13 year old child seriously injured
Damage	:	Slight hull coating damage to one boat. Major hull structural damage to the other. (Figures 2 and 3)

1.2 BACKGROUND - OFFSHORE POWERBOAT RACING

1.2.1 Adult offshore powerboat racing

Adult offshore powerboat racing is an extreme sport, where very high powered boats are raced at speeds of 100mph, or more, in open sea conditions.

The Royal Yachting Association (RYA) is the national governing body for powerboat racing in the UK. The Union Internationale Motonautique (UIM) is the international governing body. There are several classes within the overall framework of offshore powerboat racing, each separately administered by specific clubs or organisations.

All participants in powerboat racing in the UK are required to be registered and licensed by the RYA. There were about 600 RYA registered powerboaters in the UK at the time of the accident.

1.2.2 K-class offshore powerboat racing

K-class offshore powerboat racing was set up in 2002 after the organisers realised that young people were starting adult offshore racing at 16 years of age with little or no experience of handling powerboats in racing conditions. They considered this was undesirable.

K-class is the first and only junior offshore powerboat racing in the UK, and the MAIB could find no evidence of similar racing anywhere else in the world.

Wanda Offshore Racing Klubb (WORK) was set up to support K-class racing. WORK set up K-class racing to “mirror” adult offshore racing, where possible, to give the young participants as much knowledge and experience in offshore racing as feasible. Part of its objective was to make the transition from junior to senior racing safer for all concerned. It was decided that the participants should race head-to-head around a course similar, but smaller than, the design used in adult racing.(Figure 4)

All competitors in K-class racing are required to be members of WORK.

K-class racing was originally set up as one class (K-200) for drivers aged 8 to 16 years of age. However, in its first 3 years, demand continually grew, and the K-1000 sub-class was set up in 2004 for drivers aged 15 years and over. The K-400 sub-class, for drivers aged 12 years and over, was started in 2005.

The K-200 sub-class is further divided for racing purposes into two age groups, as follows:

Class A	Driver 12 to 16	Co-driver 8 to 16
Class B	Driver 8 to 12	Co-driver 8 to 16

The children race Sorcerer type powerboats in the K-200 sub-class. The boat crew consists of a driver who sits in the right-hand seat, and the co-driver in the left seat. The co-driver's job is to act as the eyes and ears of the driver and, as such, he or she will keep a careful watch during the competition for race flags and marker buoys. The co-driver also pays close attention to other competing boats as the racing proceeds, especially the boats behind, and will signal to the driver when it is clear to turn at a marker.

Before racing begins, the safety of each boat is checked by a scrutineer. During each race, adult officials are afloat in several boats. The officials include the officer of the day (OOD), a safety officer and a paramedic. Strict racing rules and protocols are applied to each K-200 race. These are very similar to those followed in the adult version of the sport.

1.3 NARRATIVE OF EVENTS

The Sorcerer 373 powerboat was originally designed in the late 1960s for use as a ski boat or beach boat. The design incorporated a low freeboard and turtle decks to facilitate the easy recovery of water skiers from the water.

Manufacture of the original GRP design began in the early 1970s, when boats were fitted with motors up to 40hp SAE. These boats were based on wooden prototypes which were fitted with 18hp SAE engines capable of towing a 90kg skier up to speeds of 25mph.

In 2002, when WORK was setting up K-class racing, it was decided to utilise Sorcerer boats, because they were readily available, could be purchased fairly cheaply and were a good size for children's racing. New boats could be supplied by a local builder, Sorcerer Ltd., who owned the design copyright, and the boats had a track history of safe racing. The new boats were called 375Rs.

The boat was also tested on the water by a member of WORK. It was found to be very stable and, at just 3.65 metres in length, an ideal size for the requirements.

The founder of WORK, who was at the time the Chairman of the RYA's Offshore Technical and UIM Working Group, informed the RYA of the club's intentions, and sought RYA backing for the venture. Taking into consideration the powerboat racing experience of the organisers, the RYA saw the establishment of K-class racing as a good and safe way to get youngsters onto the water, and to train them in the skills necessary for handling race boats. The affiliation between WORK and the RYA in K-class racing began at this time.

During the winter of 2002/3, a group of children of various ages was assembled to train for the planned 2003 K-class racing season. WORK established the minimum racing age of 8 at this time, after assessing how children of various ages coped with learning to drive the boats and understanding the rules. During the winter of 2002/2003, the children were trained, their proficiency tested, and the appropriate compulsory RYA racing licences were issued.

Some secondhand Sorcerer 373 boats were purchased and the local boatbuilder, Sorcerer Ltd, who owned the Sorcerer boat copyright, started building and supplying new racing boats, Sorcerer 375Rs, to the club and some private owners. The original boat design did not now comply with the 1998 Recreational Craft Directive, but it was still possible to build and sell such boats provided they were marked and used for racing only.

The inaugural season was planned as a series of weekend race meetings, in which children aged between 8 and 16 years of age raced against each other, head-to-head, in two age classes. Race meetings were arranged and held at various English venues, and because of the RYA affiliation the races were classified RYA National Championship Races.

On Saturday 7 June 2003, two people were sent to hospital, suspected of suffering whiplash, after the boat they were racing off Bournemouth was hit by another boat during one of the first races (see section 1.9).

Before and during this first season, concerns arose over the vulnerability of the crew in the event of a collision, and anti-intrusion bars were fitted to all Sorcerer K-200 class boats (see paragraph 1.5.1).

By the end of the first season, K200-class racing was considered by WORK and the RYA to be successful, and it was decided to continue racing the following year. In fact, another class of racing was initiated for the 2004 season, called K-1000, for persons aged 15 and above, crewing larger 4.8m (16ft) boats.

More children applied to join the club, and were trained during the winter of 2003/4 to enable them to participate in the 2004 season. By 2004, the number of Sorcerer racing boats owned privately, and by the club, had increased, and now included 7 new hulls from Sorcerer Ltd. However, a dispute between the founder of the club and Sorcerer Ltd resulted in the club ordering no more boats.

Sorcerer Ltd had concerns regarding the number of accidents that had occurred during the first racing season. They had repaired a number of damaged boats during the season, and had anecdotal evidence of the types of accidents that were occurring. The original boat designer also had concerns about K-class racing and, in particular, the accident off Bournemouth that had occurred early in the 2003 racing season. These concerns were made known to the RYA via a series of communications early in 2004.

The 2004 season started with a meeting at Allhallows Club at the end of April. The next meeting was held during May, in the docks at Bristol, a venue which is widely agreed by many powerboaters to be very demanding due to the limited space available for racing, and the close proximity of harbour walls, which tend to cause disturbed water. This was the first powerboat race meeting to be held at Bristol since racing had been stopped 14 years earlier due to safety concerns.

During the weekend racing at Bristol, two boats collided, resulting in damage to the hull of one boat by the propeller of the other boat as it had ridden up alongside her (see section 1.9).

The 2004 racing season was deemed a success by the club and the RYA and, with the number of competitors still growing, another sub-class, called K-400, was added for the 2005 season for children 12 years and older, driving 4.26m (14 ft) boats.

Training of children for all the sub-classes of racing continued during the following winter to prepare them for the 2005 season.

One of these children, Boy A, was a 9 year old who obtained an RYA licence to race dated 16 March 2005.

Two others were friends aged 11 and 13 (Boy B and Boy C), who were issued with licences dated 19 and 20 April 2005.

The first race of the 2005 season was held at Allhallows during the weekend of 23/24 April. All three boys competed, with the two friends alternating as driver and co-driver of their boat. They raced again at the next meeting in Bristol 4 weeks later where, with the younger boy, Boy B, driving, they collided with the dock wall, badly damaging the boat.

Boy A competed at the next race meeting held a few weeks later at Ramsgate on the weekend of 11/12 June, and all of the boys competed in the following meeting at Portland Harbour the next weekend.

During the afternoon of Saturday 18 June, Boys B and C were competing in a race when the driver, Boy B, pulled the boat into the centre of the course because Boy C was complaining of back pain. The child was recovered from the boat using a spinal board and transferred to a local hospital as a precautionary measure. No injury could be found during an examination at the hospital, and the boy returned to the meeting about 2 hours later. An accident reporting form was completed by WORK, and later sent to the RYA. It wrongly stated that Boy C's licence had been withdrawn.

The racing continued the following day, Sunday 19 June. At about 1315, the organisers mustered 9 boats together in a line and the Class B (younger age group) race was started. Boy B was the driver and Boy C the co-driver of boat K51, while Boy A was driving boat K11 with a co-driver who was a very experienced 13 year old boy.

The three lead boats safely passed the 1st turn marker buoy. They were being closely followed by K11 and the other 5 boats. As K11 started the turn, it suddenly stopped in the water broad-side on to the following boats. Boat K51 had been close behind and it collided with K11's port side. K51 rode up over K11's side, and K51's bow struck the 13 year old co-driver. The boat continued

climbing vertically into the air, and its two crew (Boy B and Boy C) were ejected into the sea as it turned, before it came down stern-first into the sea. K51 settled partly submerged, with just the bow showing above the water.

The race was immediately stopped (red flagged) and waterborne officials attended the scene. A paramedic in a safety boat was alerted by a course safety official to K11 after the driver (Boy A) had been seen signalling for attention. On arrival alongside, the paramedic found the co-driver lying in the boat, unconscious, with a strong pulse, but he was not breathing. The paramedic immediately started mouth to mouth ventilation, and the casualty was transferred by safety boat to the shore pontoon a short distance away.

Boy B and Boy C were retrieved uninjured from the water along with the casualty's helmet which had become dislodged during the accident.

The casualty was breathing unaided on arrival at the pontoon, and the emergency services were called by a team of St John ambulance personnel.

An ambulance arrived promptly on scene and the decision was made, due to the severity of the injuries, to transfer the casualty to hospital via the Portland coastguard rescue helicopter, based a short distance away.

The casualty was airlifted to Dorset County Hospital where the injured boy was found to be in a coma. He was later transferred to Southampton General Hospital. He recovered consciousness some days later and has continued to make a slow recovery over the succeeding weeks.

1.4 ENVIRONMENTAL CONDITIONS

On the day of the accident the environmental conditions were:

Wind	S to SE 4/5
Sea	3/4
Visibility	Very good

An RYA Commissioner's report stated:

"Choppy sea, borderline conditions at times for junior classes aged 8-12 Yrs in the K200 Sorcerer boats".

1.5 K-200 BOAT AND ENGINE TYPE

1.5.1 The boat

The K-200 race boat is based on a standard glass reinforced plastic (GRP) Sorcerer hull. There is a fore deck and windshield, behind which are two floor-mounted seats, one on runners enabling it to be adjusted fore and aft to suit the leg length of the driver. In addition, each boat is fitted with an anti-intrusion bar.

SECTION 2 - ANALYSIS

2.1 AIM

The purpose of the analysis is to determine the contributory causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents occurring in the future.

2.2 THE ACCIDENT

A spectator took a video recording of the race. This was enhanced and analysed as part of this investigation.

The collision took place at the first turn, when boat K11 violently hooked¹, and K51, which was following immediately behind and on a slightly tighter course to the marker buoy, collided with her at full speed and at a near perpendicular angle. A number of boats had been bunched together as they approached the first turn and, when K11 hooked, it virtually stopped directly in the path of K51, giving its driver no time to take avoiding action.

The precise reason that K11 hooked cannot be ascertained with any certainty. However, the three leading boats had already passed the marker buoy, and their wake, combined with the choppy seas, would have left an area of broken water at the time and location of the collision.

All class rules require the “overtaking boat” to be a minimum of two boat lengths ahead before changing course across and ahead of another boat. The children had been taught that the measurement was to be made between the bow of the leading boat and the bow of the trailing one, and this was the generally accepted interpretation.

Analysis of the video recording taken during the race showed the clear distance between the two boats to be 5.57m as they approached the first marker. In other words, more than two boat lengths bow to bow.

See Figures 12 and 13

The bow of boat K51 collided amidships with K11 at, or near, full speed. Her bow rode up over the gunwale, possibly striking the head of the co-driver of K11, before continuing to climb almost vertically into the air.

See Figure 14

¹ In powerboat racing context the term “hooking” refers to a boat inadvertently turning and stopping in a very short distance. Boats are more prone to hooking at a turn marker when the boat’s heading is already altering.

Hooking can be caused by broken water and the bow suddenly dropping into a trough, or by over steering where the driver tries to make too tight a turn often due to inexperience, or when the kill cord is inadvertently disconnected or the engine stops for another reason, leading to the bow suddenly dropping into the sea.

K11 appears to have still been moving slowly ahead as K51 continued to climb and turn, until the open hull of K51 was pointing in the direction of forward momentum, and her crew were ejected from the boat. K51 then fell stern-first back into the sea.

See Figures 15, 16, 17, 18, and 19

During the course of the collision, the helmet of K11's co-driver became dislodged, and it was later found in the sea a short distance away from the boat.

Figure 12



Position of K11 and K51 just before K11 hooked

Figure 13



Position of both boats as K11 hooked

SECTION 3 - CONCLUSIONS

3.1 SAFETY ISSUES

The following safety issues have been determined from the foregoing analysis. They are not listed in any particular order of priority.

- The collision took place at the first turn, when boat K11 violently hooked, and K51, which was following immediately behind on a slightly tighter course to the marker buoy, collided with her at full speed and a near right angle. [2.2]
- During the course of the collision, the helmet of K11's co-driver became dislodged, and it was later found in the sea a short distance away from the boat. [2.2, 2.7]
- Neither WORK nor the RYA had formally laid down minimum training criteria for new entrants who might wish to become junior powerboat racers. [2.3]
- The collision involved two young drivers who were in their first season and had minimal training in K-class racing. [2.3]
- Scrutiny of the club's records suggests that WORK was not giving sufficient priority to the completion and submission of accident and incident reports. [2.4]
- The accident reports that were received by the RYA received only minimal attention. [2.4]
- K-class races were very well organised, and several volunteer adult race officials were involved, many of whom were stationed around the course in boats during the racing. [2.5]
- When the accident occurred, the paramedic was on scene in seconds and able to administer the necessary lifesaving first-aid. Had this not been the case, the injuries sustained and the outcome might have been worse. [2.5.1]
- When choosing the course design for Junior Powerboat racing, it appears that the desire to make it similar to senior racing outweighed any concerns that there might have been about boat safety at the first turn. [2.5.2]
- Anti-intrusion bars were fitted to all K-200 class boats in 2003 as a result of some accidents and incidents. However, they were not sufficiently robust to protect them in the case of a right angled or near right angled collision between two boats. [2.5.3]
- Warnings given to the RYA by the boat's designer were not heeded. [2.5.4]
- The proposed change from 20hp to 25hp engines should have been the subject of a thorough risk assessment before the rules for K-class racing were changed. [2.5.5]

- When K-class racing was set up, it was not subjected to a rigorous risk assessment process to ensure the children's safety. Such a risk assessment would have recognised the particular problems involved at the first turn, and the need for counter-measures to bring the risk within acceptable limits would have been identified. [2.5.7]
- The RYA should have ensured WORK's risk assessments were accurate and effective before giving approval for the new class, and thereafter they should have closely monitored all developments within the sport. [2.6]
- There has never been any proper assessment of the requirements of a safety helmet for powerboat racing which would lead to the identification, or development, of a suitable standard. [2.7]
- The MK10 GMSH was only tested for sizes 57cm to 62cm. As such, it was not tested for safety for many head sizes, including most children's. [2.7.1]
- What caused the helmet of K11's co-pilot to come off during the accident may never be established. However, the boy's head was smaller and outside the range of head sizes for which the helmet had been tested, and an important part of the helmet's securing arrangement was missing. [2.7.1]
- The development of K-class racing seems to have lacked a detached and objective oversight bearing in mind the potential risk to the children posed by competitive racing. [2.8]